Numerical Integration and LTI System Stability

Problem 1.2.19 (EE 2007)

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Problem Statement

The differential equation $\frac{dx}{dt} = \frac{1-x}{\tau}$ is discretised using Euler's numerical integration method with a time step $\Delta T > 0$.
What is the maximum permissible value of ΔT to ensure stability of the solution of the corresponding discrete time equation?

a) 1 b) $\frac{\tau}{2}$ c) τ d) 2τ

Note: The problem image contains a typographical error ($\frac{dx}{dy}$ instead of $\frac{dx}{dt}$), but explicitly defines ΔT as the time step. We proceed using standard time derivative notation.

Euler's Forward Discretization

We begin with the fundamental definition of the forward difference approximation for the derivative:

$$\frac{dx}{dt} \approx \frac{x[n+1] - x[n]}{\Delta T} \tag{1}$$

Substituting this into our given differential equation:

$$\frac{x[n+1] - x[n]}{\Delta T} = \frac{1 - x[n]}{\tau} \tag{2}$$

Rearranging to form a linear first-order difference equation (a discrete-time LTI system):

$$x[n+1] = x[n] + \frac{\Delta T}{\tau}(1 - x[n]) \tag{3}$$

$$x[n+1] = \left(1 - \frac{\Delta T}{\tau}\right) x[n] + \frac{\Delta T}{\tau} \tag{4}$$

Z-Transform and System Pole

To rigorously analyze stability, we take the Z-transform of the homogeneous part of the difference equation, $x[n+1] - \left(1 - \frac{\Delta T}{\tau}\right) x[n] = 0$:

$$zX(z) - \left(1 - \frac{\Delta T}{\tau}\right) X(z) = 0 \tag{5}$$

The characteristic equation dictates the dynamics of the system:

$$z - \left(1 - \frac{\Delta T}{\tau}\right) = 0 \tag{6}$$

This yields a single system pole at:

$$p = 1 - \frac{\Delta T}{\tau} \tag{7}$$

The Unit Circle Stability Criterion

For any discrete-time LTI system to be asymptotically stable, its poles must lie strictly inside the unit circle on the Z-plane.

$$|p| < 1 \tag{8}$$

Substituting our pole location into the stability criterion:

$$\left|1 - \frac{\Delta T}{\tau}\right| < 1 \tag{9}$$

Expanding the absolute value inequality:

$$-1 < 1 - \frac{\Delta T}{\tau} < 1 \tag{10}$$

Solving for Maximum Time Step

Subtract 1 from all parts of the inequality:

$$-2 < -\frac{\Delta T}{\tau} < 0 \tag{11}$$

Multiply by $-\tau$ (Note: multiplying by a negative flips the inequality signs, assuming $\tau > 0$ for a physical system):

$$2\tau > \Delta T > 0 \tag{12}$$

$$0 < \Delta T < 2\tau \tag{13}$$

The strict limit for asymptotic stability is 2τ. If ΔT reaches or exceeds this, the discrete sequence diverges.

Correct Option: (d) 2τ

Python Computational Verification

To verify this boundary, we simulate the difference equation natively in Python across three distinct stability regimes.

```
import numpy as np
import matplotlib.pyplot as plt

def simulate_euler(tau, dt, steps, x0=0):
    x = np.zeros(steps)
    x[0] = x0
    for n in range(steps - 1):
        x[n+1] = (1 - dt/tau)*x[n] + dt/tau
    return x

tau = 1.0   # Set arbitrary time constant
steps = 25
n_axis = np.arange(steps)

# 1. Perfectly Stable (Pole at 0)
x_stable = simulate_euler(tau, dt=1.0, steps=steps)
# 2. Marginally Stable / Ringing (Pole at -0.9)
x_ringing = simulate_euler(tau, dt=1.9, steps=steps)
# 3. Unstable / Divergent (Pole at -1.1)
x_unstable = simulate_euler(tau, dt=2.1, steps=steps)

plt.figure(figsize=(8, 4.5))
plt.plot(n_axis, x_stable, 'b-o', label=r'$\Delta T = \tau$ (Stable)')
plt.plot(n_axis, x_ringing, 'g-s', label=r'$\Delta T = 1.9\tau$ (Ringing)')
plt.plot(n_axis, x_unstable, 'r-^', label=r'$\Delta T = 2.1\tau$ (Unstable)')

plt.axhline(1.0, color='k', linestyle='--', label='Theoretical Steady State')
plt.ylim(-1, 3)
```

Simulation Results

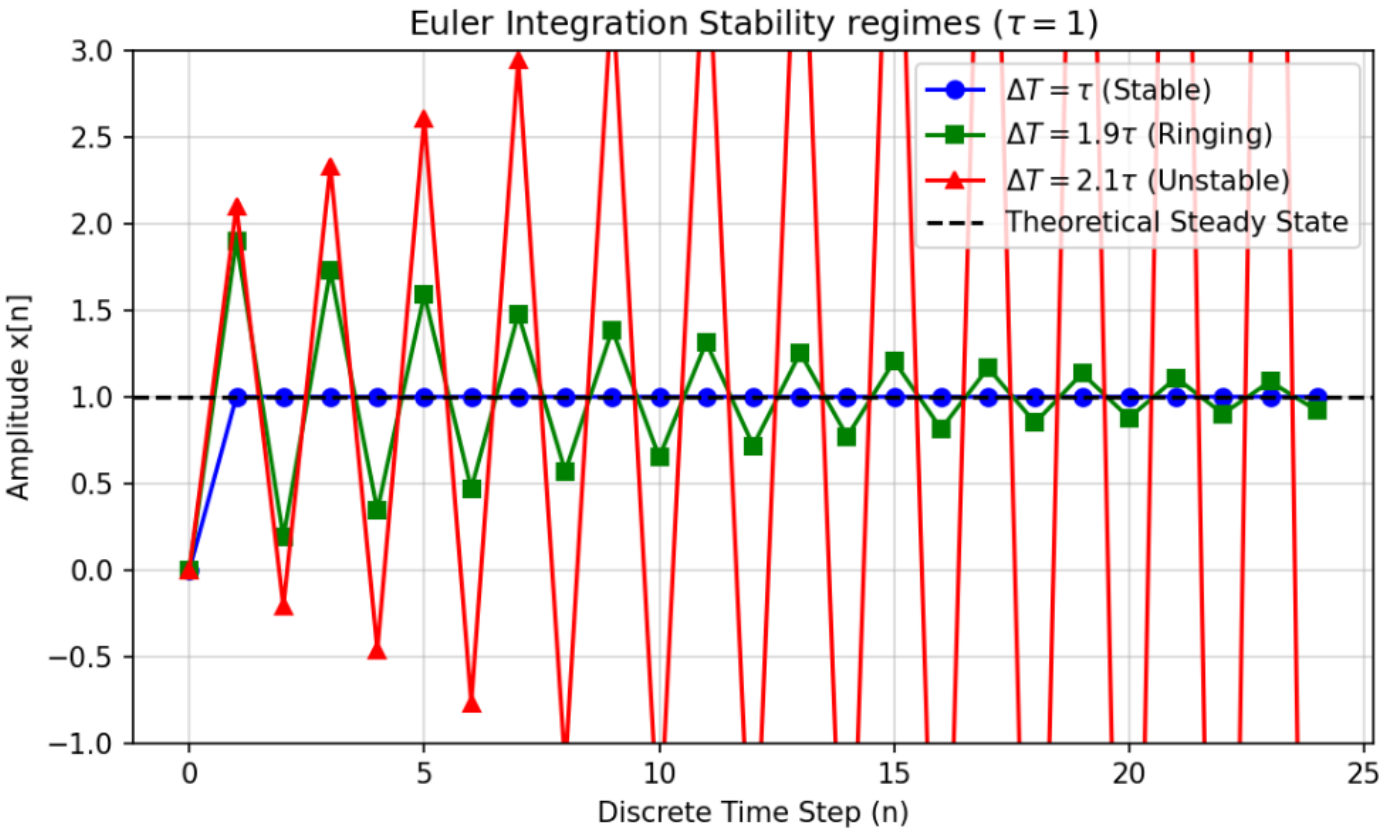

Figure: Computational proof of the 2τ boundary. Notice the severe oscillation (ringing) as ΔT approaches 2τ, and immediate divergence when it exceeds it.